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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
23CSE302 – COMPUTER NETWORKS

EXPERIMENT 4

IMPLEMENTATION OF DHCP SERVER

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Date of Exp: 13-07-2026	Date of Submission:	

OBJECTIVES:

- 1. Students will gain indepth understanding of how to setup a Server as a DHCP Server**
- 2. Students will have exposure to configure a Router as a DHCP Server**

A. IMPLEMENTATION OF A BASIC DHCP SERVER ON A SINGLE DEVICE

AIM

To configure a DHCP server on a PC-PT device using Cisco Packet Tracer and verify that the client device can obtain an IP address automatically.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 1) End Devices
 - PC-PT
 - Server -PT
- 2) Connections
 - Copper Cross-over

PROCEDURE

To configure a DHCP server on a network using Cisco Packet Tracer and obtain an IP address automatically, follow these steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add one PC:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag a PC into the workspace.
- 2) Add a server:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "Server" and drag a server into the workspace.

Step 3: Connect the Devices

- 1) Select the Connection Tool:
 - Click on the "Connections" icon (lightning bolt) in bottom left corner.
- 2) Connect the PC to the server:
 - Click on a PC and then click on the server to connect them with a Copper Cross-Over cable.

Step 4: IP Configuration for Server-PT

- 1) Open the Server, navigate to Config, and Edit FastEthernet0 tab.
- 2) Choose Static to set IP address and subnet mask,

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

- 3) Check the box for "Port Status" to bring the interface up.

Step 5: Configure the DHCP Server:

- 1) Click on the Server-PT device to open its configuration window.
- 2) Navigate to the Services tab and Select DHCP from the services list.
- 3) Turn on the DHCP service if it is not already on.
- 4) Configure the DHCP settings:
 - Set the Start IP Address to **192.168.1.2**
 - Enter the Subnet Mask as **255.255.255.0**
 - Enter Maximum Number of Users as **3**
 - Click on Add to save the DHCP pool.

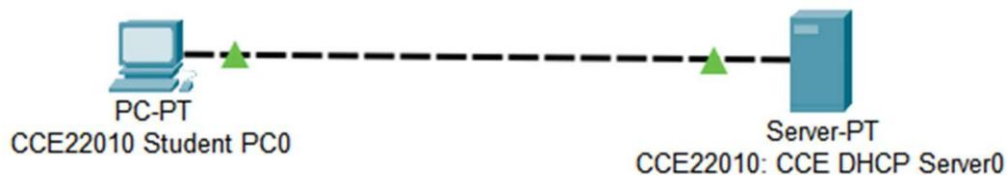
Step 6: Configure the PC-PT to Use DHCP:

- 1) Click on the PC-PT to open its configuration window.
- 2) Navigate to the Desktop tab and then to IP Configuration.
- 3) Select the DHCP option to obtain an IP address automatically via DHCP.

Step 7: Verify the Configuration:

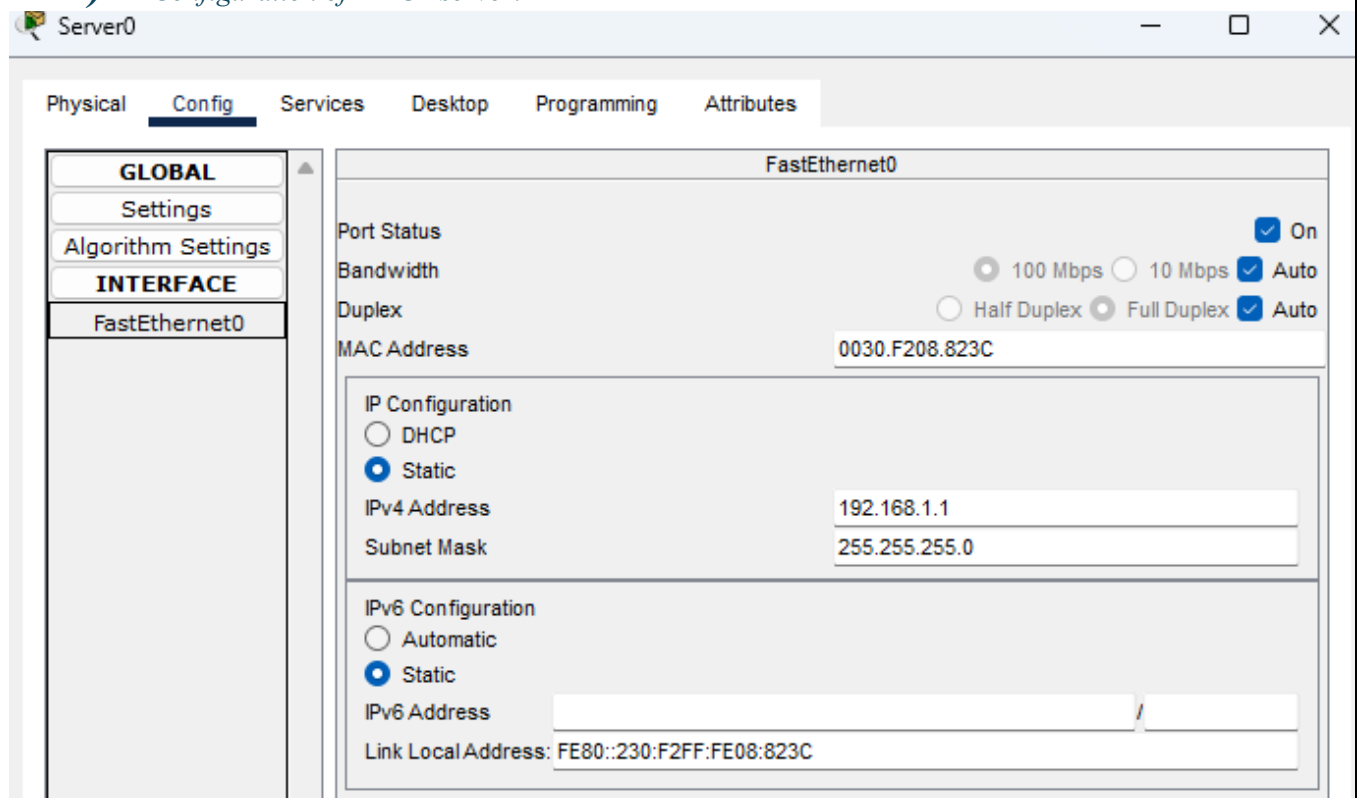
- 1) Return to the Desktop tab of the PC-PT and open the Command Prompt.
- 2) Type **ipconfig** and press Enter to see the IP address assigned by the DHCP server.
- 3) Type **ipconfig /release** to release the IP address back to the DHCP pool, disconnect from the network and prepare for obtaining a new IP address.
- 4) Type **ipconfig /renew** to re-establish network connectivity and get a new IP address after releasing the previous one.

NETWORK TOPOLOGY DIAGRAM



OUTPUT

1) IP Configuration of DHCP server:



2)Configure the DHCP Server:

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.10

Start IP Address : 192 168 1 0

Subnet Mask: 255 255 255 0

Maximum Number of Users : 3

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	192.168....	192.168....	255.255....	3	0.0.0.0	0.0.0.0

2) *Configure the PC-PT to Use DHCP:*

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 192.168.1.10

IPv6 Configuration

3) Verify the Configuration:

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::290:2BFF:FEC6:69A8
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.1.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

```
C:\>ipconfig /release

IP Address. . . . .: 0.0.0.0
Subnet Mask . . . . .: 0.0.0.0
Default Gateway . . . . .: 0.0.0.0
DNS Server . . . . .: 0.0.0.0

C:\>
```

```
C:\>ipconfig /renew

IP Address. . . . .: 192.168.1.2
Subnet Mask . . . . .: 255.255.255.0
Default Gateway . . . . .: 192.168.1.1
DNS Server . . . . .: 192.168.1.10

C:\>
```

RESULT

1. First IP Configuration: obtain an IP address automatically via DHCP for Faculty PC.

The PC successfully obtained an IP address automatically from the DHCP server.

2. Release of IP address back to the DHCP pool using **ipconfig /release** and re-establish network connectivity to get a new IP address using **ipconfig /renew**:
The IP address was successfully released using **ipconfig /release** and renewed using **ipconfig /renew**.

3. The new IP address is auto-updated in the IP configuration tab of Faculty PC:

The PC received a valid IP address automatically after renewal, confirming successful DHCP operation.

INFERENCE

The DHCP server successfully assigned IP addresses automatically to the client PC. This reduced manual IP configuration and ensured proper communication between the server and the client, demonstrating the efficient working of the DHCP protocol.

B. IMPLEMENTATION OF A DHCP SERVER ON DEVICES IN THE SAME NETWORK

AIM

To configure a DHCP server on a network using Cisco Packet Tracer and verify that the client device can obtain an IP address automatically.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 1) End Devices
 - 5 PC-PTS
 - Server -PT
- 2) Network Devices
 - Switch 2950-24
- 3) Connections
 - Copper Straight-Through

PROCEDURE

To configure a DHCP server on a network using Cisco Packet Tracer and obtain an IP address automatically, follow these steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add four PCs:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag four PCs into the workspace.
- 2) Add a server:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "Server" and drag a server into the workspace.

3) Add a Switch:

- Click on the "Switches" icon in the bottom left corner.
- Select a switch (2950-24) and drag it into the workspace.

Step 3: Connect the Devices

1) Select the Connection Tool:

- Click on the "Connections" icon (lightning bolt) in bottom left corner.

2) Connect the PCs to the Switch:

- Click on a PC and then click on the switch to connect them with a copper straight-through cable.
- Repeat the process for the four PCs.

3) Connect the Switch to the Server:

- Click on the switch and then click on the server to connect them with a copper straight-through cable.

Step 4: IP Configuration for Server-PT

1) Open the Server, navigate to Config, and Edit FastEthernet0 tab.

2) Choose Static to set IP address and subnet mask,

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

3) Check the box for "Port Status" to bring the interface up.

Step 5: Configure the DHCP Server:

1) Click on the Server-PT device to open its configuration window.

2) Navigate to the Services tab and Select DHCP from the services list.

3) Turn on the DHCP service if it is not already on.

4) Configure the DHCP settings:

- Set the Start IP Address to **192.168.1.2**
- Enter the Subnet Mask as **255.255.255.0**
- Enter Maximum Number of Users as **4**
- Click on Add to save the DHCP pool.

Step 6: Configure the PC-PT to Use DHCP:

1) Click on the PC-PT to open its configuration window.

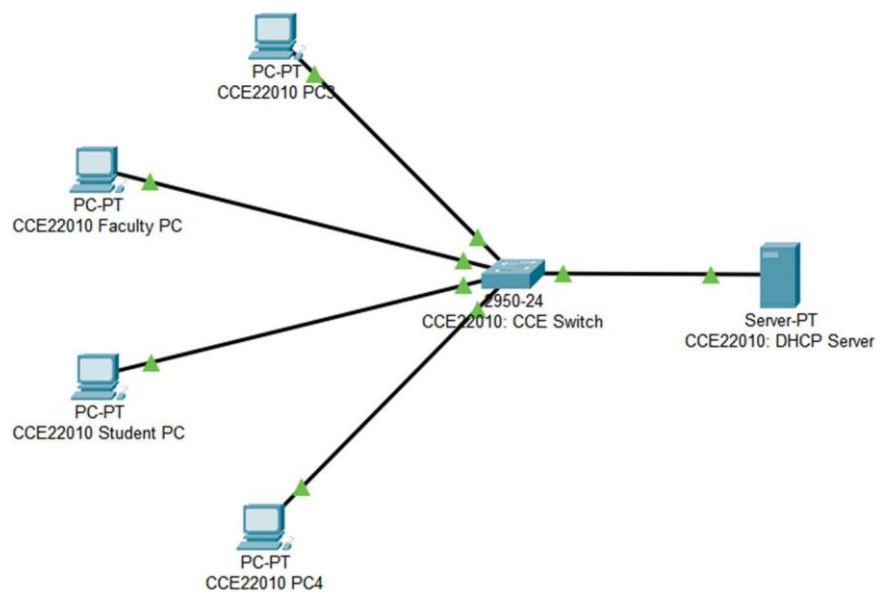
2) Navigate to the Desktop tab and then to IP Configuration.

3) Select the DHCP option to obtain an IP address automatically via DHCP.

Step 6: Verify the Configuration:

- 1) Return to the Desktop tab of the PC-PT and open the Command Prompt.
- 2) Type **ipconfig** and press Enter to see the IP address assigned by the DHCP server.
- 3) Type **ipconfig /release** to release the IP address back to the DHCP pool, disconnect from the network and prepare for obtaining a new IP address.
- 4) Type **ipconfig /renew** to re-establish network connectivity and get a new IP address after releasing the previous one.

NETWORK TOPOLOGY DIAGRAM



OUTPUT

1) IP Configuration of DHCP server:

The screenshot shows the configuration window for 'Server0'. The 'Config' tab is selected, and the 'FastEthernet0' interface is chosen under the 'INTERFACE' section. The 'Port Status' is set to 'On'. The 'Bandwidth' is set to '100 Mbps'. The 'Duplex' is set to 'Full Duplex'. The 'MAC Address' is '000B.BE8B.68E7'. The 'IP Configuration' is set to 'Static' with an 'IPv4 Address' of '192.168.1.1' and a 'Subnet Mask' of '255.255.255.0'. The 'IPv6 Configuration' is set to 'Static' with an 'IPv6 Address' field and a 'Link Local Address' of 'FE80::20B:BEFF:FE8B:68E7'.

Server0

Physical **Config** Services Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

FastEthernet0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000B.BE8B.68E7

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::20B:BEFF:FE8B:68E7

2) Configure the DHCP Server:

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.10

Start IP Address: 192 168 1 2

Subnet Mask: 255 255 255 0

Maximum Number of Users: 4

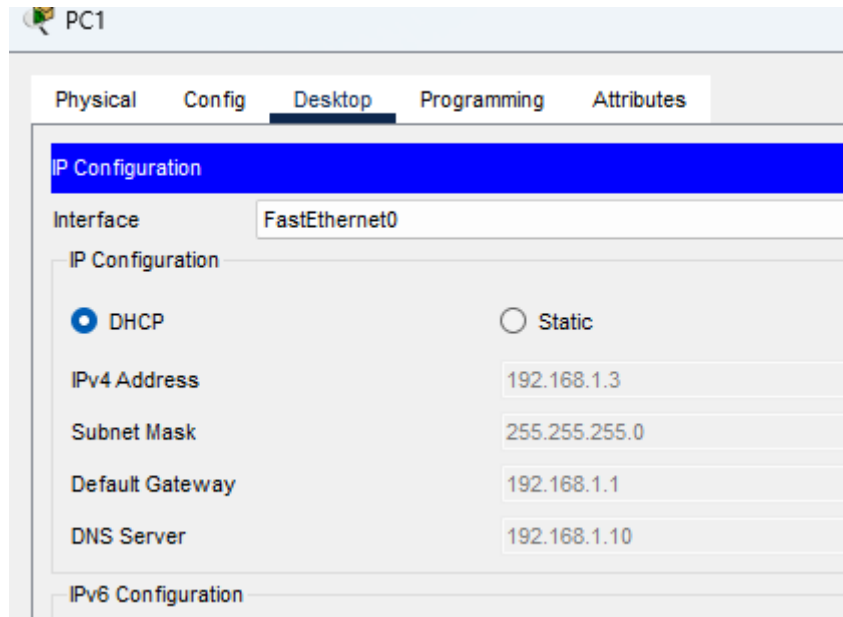
TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	192.168....	192.168....	255.255....	4	0.0.0.0	0.0.0.0

3) Configure the PC-PTs to Use DHCP:



The screenshot shows the configuration window for PC1. The 'Desktop' tab is selected, and the 'IP Configuration' section is highlighted. The interface is set to 'FastEthernet0'. Under 'IP Configuration', the 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The following fields are visible:

Field	Value
IPv4 Address	192.168.1.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.10

Below the IPv4 Configuration section, the 'IPv6 Configuration' section is visible but currently empty.

4) Verify the Configuration:

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::2E0:F7FF:FED5:680D
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.1.4
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                           192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                           0.0.0.0
```

```
C:\>ipconfig /release

IP Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: 0.0.0.0
DNS Server.....: 0.0.0.0

C:\>ipconfig /renew

IP Address.....: 192.168.1.4
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.1.1
DNS Server.....: 192.168.1.10

C:\>
```

RESULT

1. First IP Configuration: obtain an IP address automatically via DHCP for Faculty PC.

The client PCs successfully obtained IP addresses automatically from the DHCP server.

2. Release of IP address back to the DHCP pool using **ipconfig /release** and re-establish network connectivity to get a new IP address using **ipconfig/renew**:

The IP address was successfully released using **ipconfig /release** and renewed using **ipconfig /renew**.

INFERENCE:

The DHCP server successfully assigned dynamic IP addresses to multiple client PCs connected through the switch. This demonstrated centralized and automatic IP address allocation within the same network.

C. CONFIGURING ROUTER AS DHCP SERVER FOR ONE END DEVICE

AIM

To configure a Cisco router as a DHCP server for a PC-PT device using Cisco Packet Tracer and verify that it assigns IP addresses dynamically to the connected devices.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 1) End Devices
 - PC-PT
- 2) Network Devices
 - Router 2911
- 3) Connections
 - Copper Straight-Through

PROCEDURE

To configure a Cisco router as a DHCP server on a network using Cisco Packet Tracer and obtain an IP address automatically, follow these steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add one PCs:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag a PC into the workspace.
- 2) Add a Router:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select " Router " and drag a router (2901 series) into the workspace.

Step 3: Connect the Devices

- 1) Select the Connection Tool:
 - Click on the "Connections" icon (lightning bolt) in bottom left corner.
- 2) Connect the PC to the router:
 - Click on a PC and then click on the router to connect them with a Copper Straight-Through cable.

Step 4: Configure IP Address of Router

- 1) Click on the router and navigate to the Config tab.
- 2) Select the interface connected to the PC (e.g., GigabitEthernet0/0).
- 3) Set the IP address and subnet mask:
IP Address: 192.168.1.1
Subnet Mask: 255.255.255.0
- 4) Check the box for "Port Status" to bring the interface up.

Step 5: Configure the Router as DHCP Through Command Line Interface

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Exclude the specified range of IP addresses (192.168.1.1 to 192.168.1.5) from being assigned by the DHCP server:
ip dhcp excluded-address 192.168.1.1 192.168.1.5
- 5) Create a new DHCP pool named "pool1" and enter DHCP pool configuration mode:
ip dhcp pool pool1
- 6) Define the network range for the DHCP pool.
network 192.168.1.0 255.255.255.0
- 7) Specify the DNS server IP address (192.168.1.10) that will be provided to DHCP clients.
dns-server 192.168.1.10
- 8) Sets the default gateway (router) IP address (192.168.1.1) for DHCP clients:
default-router 192.168.1.1
- 9) Exit the DHCP pool configuration mode
exit

Step 6: Verify the Configuration and obtain IP address for the PC

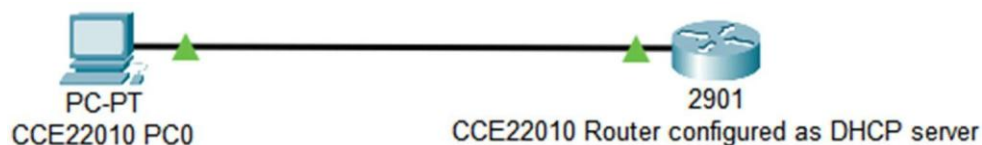
Through graphical-user interface:

- 1) On the PC, go to the Desktop tab, open the IP Configuration, and set it to DHCP.
- 2) Verify that the PC obtains an IP address from the router.

Through Command Line Interface:

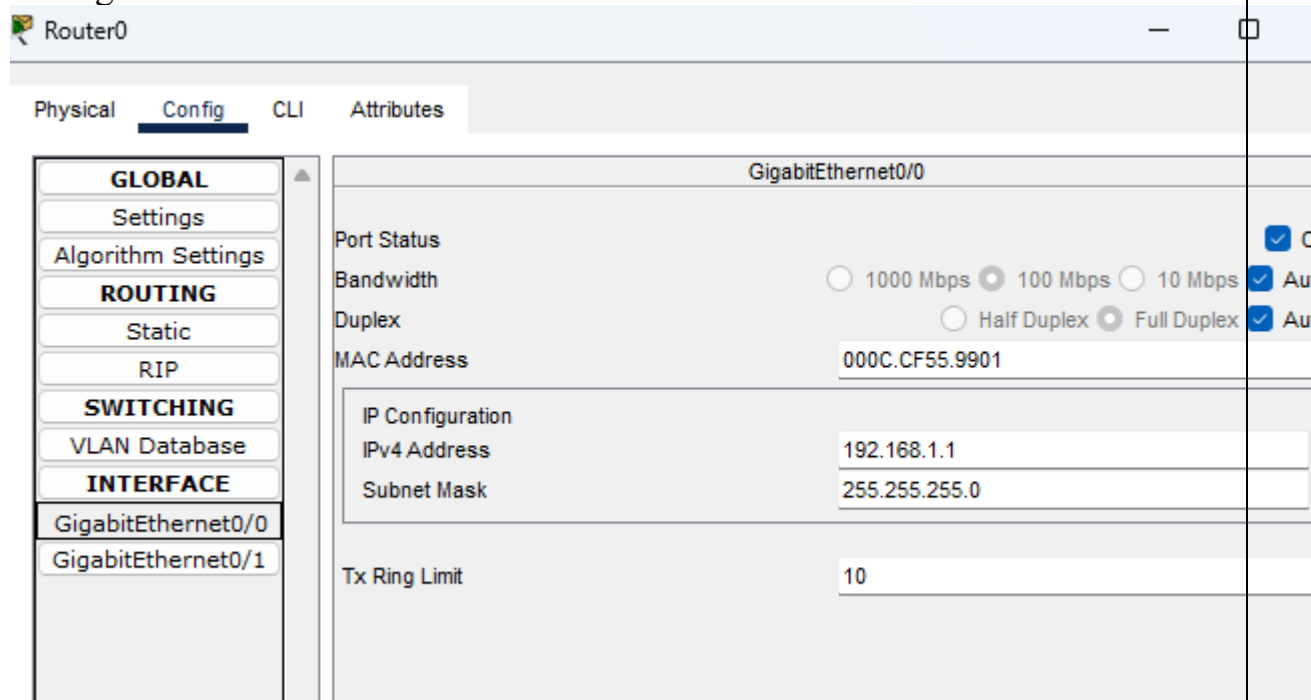
- 1) Type **ipconfig /release** to release the IP address back to the DHCP pool, disconnect from the network and prepare for obtaining a new IP address.
- 2) Type **ipconfig /renew** to re-establish network connectivity and get a new IP address from the DHCP pool after releasing the previous one.

NETWORK TOPOLOGY DIAGRAM



OUTPUT

- 1) Configure IP Address of Router:



2) Configure the Router as DHCP Through Command Line Interface

```
^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.5
Router(config)#ip dhcp pool pool1
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#dns-server 192.168.1.10
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#exit
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

3) Verify Configuration:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::290:21FF:FE3E:4919
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.1.6
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ipconfig /release

    IP Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0
    DNS Server . . . . .: 0.0.0.0

C:\>ipconfig /renew

    IP Address. . . . .: 192.168.1.6
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.1.1
    DNS Server . . . . .: 192.168.1.10

C:\>
```

RESULT

1. First IP Configuration: obtain an IP address automatically via DHCP for Faculty PC.

The PC successfully obtained an IP address automatically from the router configured as a DHCP server.

2. Release of IP address back to the DHCP pool using **ipconfig /release** and re-establish network connectivity to get a new IP address using **ipconfig /renew**:

The IP address was successfully released using **ipconfig /release** and renewed using **ipconfig /renew**.

3. The new IP address is auto-updated in the IP configuration tab of Faculty PC: After renewal, the PC automatically received a valid IP address from the router's DHCP pool.

INFERENCE

The Cisco router was successfully configured as a DHCP server and dynamically assigned IP addresses to the connected PC. This eliminated the need for manual IP configuration and demonstrated efficient IP address management using DHCP.

D. CONFIGURING ROUTER AS DHCP SERVER ON DEVICES IN THE SAME NETWORK

AIM

To configure a Cisco router as a DHCP server on a network using Cisco Packet Tracer and verify that it assigns IP addresses dynamically to the connected devices.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 1) End Devices
 - 5 PC-PT
- 2) Network Devices
 - Router 2911
 - Switch 2950-24
- 3) Connections
 - Copper Straight-Through
 - Console cable

PROCEDURE

To configure a Cisco router as a DHCP server on a network using Cisco Packet Tracer and obtain an IP address automatically, follow these steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add Five PCs:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag five PCs into the workspace.

- 2) Add a Router:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select " Router " and drag a router (2901 series) into the workspace.
- 3) Add a Switch:
 - Click on the "Switches" icon in the bottom left corner.
 - Select a switch (2950-24) and drag it into the workspace.

Step 3: Connect the Devices

- 1) Select the Connection Tool:
 - Click on the "Connections" icon (lightning bolt) in bottom left corner.
- 2) Connect the PC to the switch:
 - Click on a PC and then click on the switch to connect them with a Copper Straight-Through cable.
 - Repeat the process for the five PCs.
- 3) Connect the Switch to the Router:
 - Click on the switch and then click on the router to connect them with a copper straight-through cable.
- 4) Connect a PC to the router:
 - Click on a PC and then click on the router to connect them with a console cable.

Step 4: Configure IP Address of Router

- 1) Click on the router and navigate to the Config tab.
- 2) Select the interface connected to the PC (e.g., GigabitEthernet0/0).
- 3) Set the IP address and subnet mask:
 - IP Address: 192.168.1.1*
 - Subnet Mask: 255.255.255.0**
- 4) Check the box for "Port Status" to bring the interface up.

Step 5: Configure the Router as DHCP Through Command Line Interface

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
 - enable*

- 3) Enter global configuration mode by typing:
configure terminal
- 4) Exclude the specified range of IP addresses (192.168.1.1 to 192.168.1.5) from being assigned by the DHCP server:
ip dhcp excluded-address 192.168.1.1 192.168.1.5
- 5) Create a new DHCP pool named "pool1" and enter DHCP pool configuration mode:
ip dhcp pool pool1
- 6) Define the network range for the DHCP pool.
network 192.168.1.0 255.255.255.0
- 7) Specify the DNS server IP address (192.168.1.10) that will be provided to DHCP clients.
dns-server 192.168.1.10
- 8) Sets the default gateway (router) IP address (192.168.1.1) for DHCP clients:
default-router 192.168.1.1
- 9) Exit the DHCP pool configuration mode
Exit

Step 6: Verify the Configuration and obtain IP address for the PC

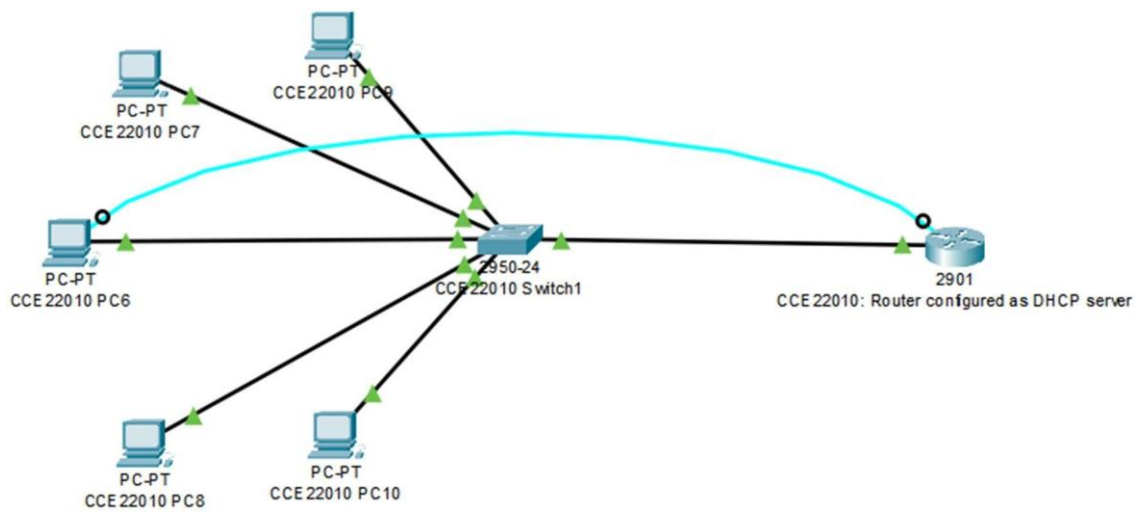
Through graphical-user interface:

- 3) On the PC, go to the Desktop tab, open the IP Configuration, and set it to DHCP.
- 4) Verify that the PC obtains an IP address from the router.

Through Command Line Interface:

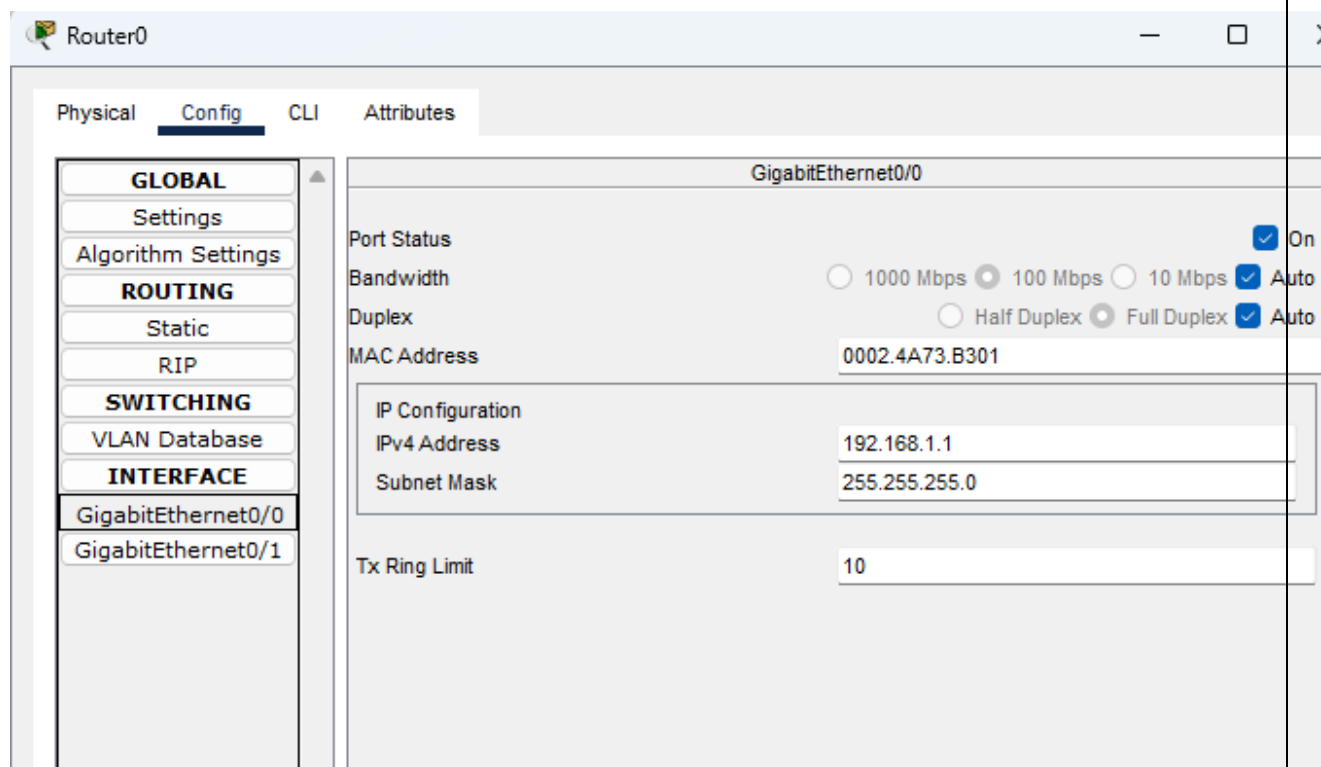
- 3) Type **ipconfig /release** to release the IP address back to the DHCP pool, disconnect from the network and prepare for obtaining a new IP address.
- 4) Type **ipconfig /renew** to re-establish network connectivity and get a new IP address from the DHCP pool after releasing the previous one.

NETWORK TOPOLOGY DIAGRAM



OUTPUT

1) Configure IP Address of Router:

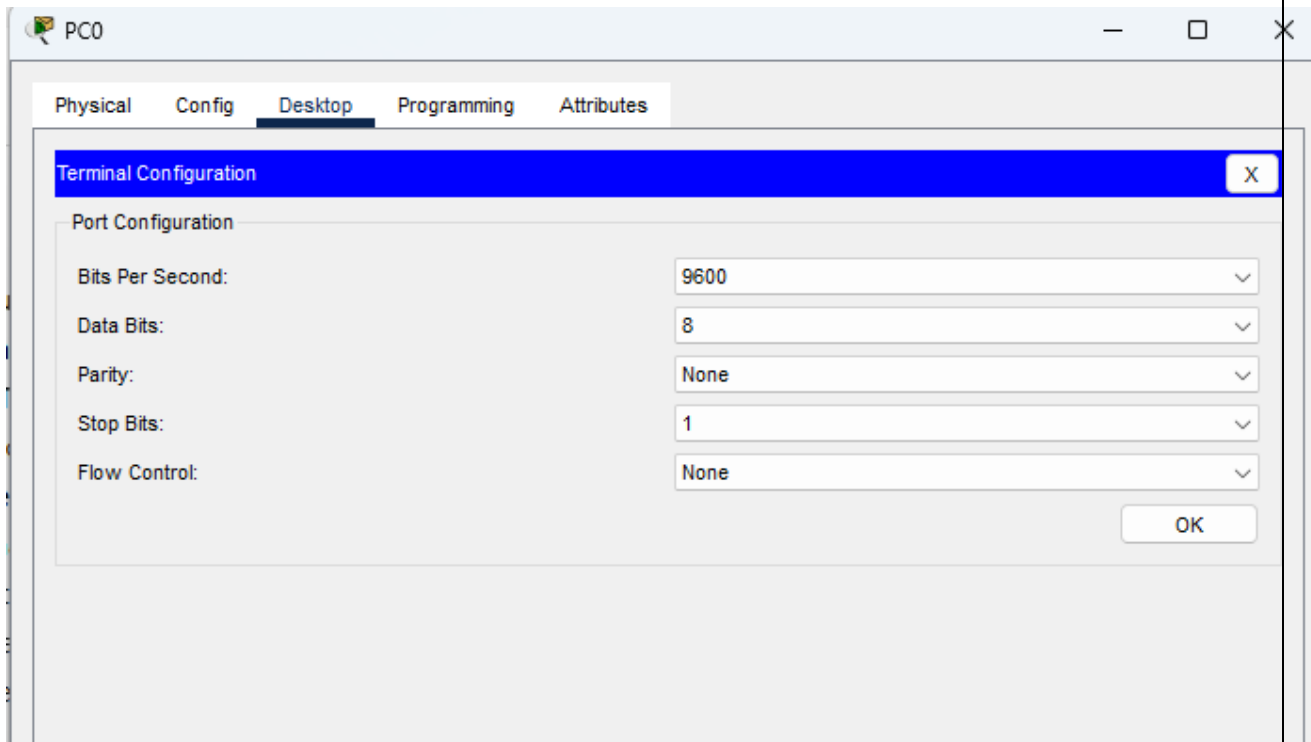


2) Configure the Router as DHCP Through Command Line Interface

* invalid input detected at marker.

```
Router(config-if)#ip dhcp excluded-address 192.168.1.1 192.168.1.5
Router(config)#ip dhcp pool pool1
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#dns-server 192.168.1.10
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#exit
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

3) Access and Configure the Router from PC terminal using console cable:



4) Verify Configuration through any PC:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:8FFF:FE64:CE44
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.1.6
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ipconfig /release

    IP Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0
    DNS Server . . . . .: 0.0.0.0

C:\>ipconfig /renew

    IP Address. . . . .: 192.168.1.6
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.1.1
    DNS Server . . . . .: 192.168.1.10

C:\>
```

RESULT

We successfully configured a Cisco router to act as a DHCP server. The router dynamically assigned IP addresses to the connected devices within the defined IP range, ensuring efficient IP address management in the network.

1. First IP Configuration: obtain an IP address automatically via DHCP for Faculty PC.

The PC successfully obtained an IP address automatically from the router configured as a DHCP server.

2. Release of IP address back to the DHCP pool using **ipconfig /release** and re-establish network connectivity to get a new IP address using **ipconfig /renew**:

The IP address was successfully released using `ipconfig /release` and renewed using `ipconfig /renew`.

After renewal, the PC automatically received a valid IP address from the router's DHCP pool.

INFERENCE

The Cisco router was successfully configured as a DHCP server and dynamically assigned IP addresses to the connected PC. This eliminated the need for manual IP configuration and demonstrated efficient IP address management using DHCP.

E. CONFIGURATION OF ROUTER AS DHCP SERVER AND IMPLEMENTATION OF A WEB SERVER IN A NETWORK

AIM

To configure a DHCP server on a Cisco router for assigning IP addresses dynamically to the connected devices and to implement a web server to access a webpage from connected devices within the network using Cisco Packet Tracer.

SYSTEM AND SOFTWARE REQUIREMENTS

Software – Cisco Packet Tracer

Components used:

- 4) End Devices
 - 5 PC-PT
 - Server-PT
- 5) Network Devices
 - Router 2911
 - Switch 2950-24
- 6) Connections
 - Copper Straight-Through
 - Console cable

PROCEDURE

To configure a Cisco router as a DHCP server on a network using Cisco Packet Tracer and obtain an IP address automatically and to implement a web server to access a webpage from connected devices within the network, follow these steps:

Step 1: Open Cisco Packet Tracer

- 1) Launch Cisco Packet Tracer on your computer.

Step 2: Add Devices to the Workspace

- 1) Add Five PCs:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "PC" and drag five PCs into the workspace.
- 2) Add a server:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "Server" and drag a server into the workspace.
- 3) Add a Router:
 - Click on the "End Devices" icon in the bottom left corner.
 - Select "Router" and drag a router (2901 series) into the workspace.
- 4) Add a Switch:
 - Click on the "Switches" icon in the bottom left corner.
 - Select a switch (2950-24) and drag it into the workspace.

Step 3: Connect the Devices

- 1) Select the Connection Tool:
 - Click on the "Connections" icon (lightning bolt) in bottom left corner.
- 2) Connect the PC to the switch:
 - Click on a PC and then click on the switch to connect them with a Copper Straight-Through cable.
 - Repeat the process for the five PCs.
- 3) Connect the Server to the switch:
 - Click on a Server and then click on the switch to connect them with a Copper Straight-Through cable.
- 4) Connect the Switch to the Router:
 - Click on the switch and then click on the router to connect them with a copper straight-through cable.
- 5) Connect a PC to the router:
 - Click on a PC and then click on the router to connect them with a console cable.

Step 4: IP Configuration

- 1) Router IP configuration:
 - Click on the router and navigate to the Config tab.
 - Select the interface connected to the PC (e.g., GigabitEthernet0/0).
 - Set the IP address and subnet mask:
IP Address: 192.168.1.1
Subnet Mask: 255.255.255.0
 - Check the box for "Port Status" to bring the interface up.

2) Server IP configuration:

- Click on the Server and navigate to the Config tab.
- Select the interface connected to the PC (e.g., FastEthernet0/0).
- Select Static configuration and Set the IP address and subnet mask:
IP Address: 192.168.1.2
Subnet Mask: 255.0.0.0
- Check the box for "Port Status" to bring the interface up.

Step 5: Configure the Router as DHCP Through Command Line Interface:

- 1) Select the router, navigate to CLI
- 2) Enter privileged EXEC mode by typing:
enable
- 3) Enter global configuration mode by typing:
configure terminal
- 4) Exclude the specified range of IP addresses (192.168.1.1 to 192.168.1.5) from being assigned by the DHCP server:
ip dhcp excluded-address 192.168.1.1 192.168.1.5
- 5) Create a new DHCP pool named "pool1" and enter DHCP pool configuration mode:
ip dhcp pool pool1
- 6) Define the network range for the DHCP pool.
network 192.168.1.0 255.255.255.0
- 7) Specify the DNS server IP address (192.168.1.10) that will be provided to DHCP clients.
dns-server 192.168.1.10
- 8) Sets the default gateway (router) IP address (192.168.1.1) for DHCP clients:
default-router 192.168.1.1
- 9) Exit the DHCP pool configuration mode
Exit

Step 6: Web Server Configuration:

- 1) Click on the Server-PT device to open its configuration window.
- 2) Navigate to Services, open HTTP, and ensure the service is ON.
- 3) Make any necessary changes to the webpage by editing the index.html.
- 4) Navigate to the Config tab and Edit the global settings
- 5) Name the server (ex: Amrita Web Server)

6) Configure the Gateway DNS IPv4 settings:

- Select Static configuration and set the default gateway and DNS server IP address as follows,

Default Gateway: 192.168.1.2

DNS Server: 192.168.1.10

Step 7: Verify the Configuration and obtain IP address for the PC Through graphical-user interface:

- 1) On the PC, go to the Desktop tab, open the IP Configuration, and set it to DHCP.
- 2) Verify that the PC obtains an IP address from the router.

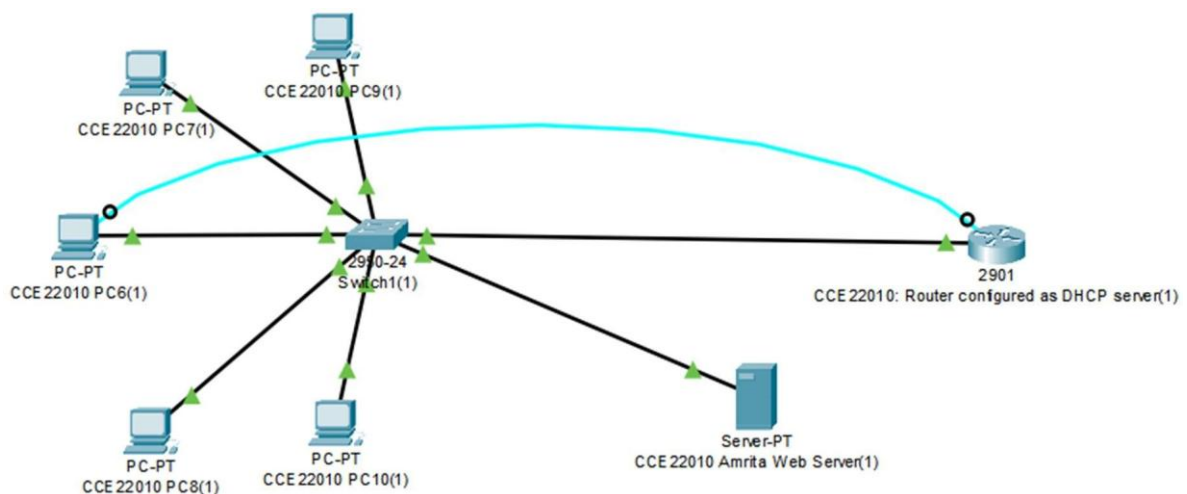
Through Command Line Interface:

- 1) Type **ipconfig /release** to release the IP address back to the DHCP pool, disconnect from the network and prepare for obtaining a new IP address.
- 2) Type **ipconfig /renew** to re-establish network connectivity and get a new IP address from the DHCP pool after releasing the previous one.

Step 8: Verify the web service

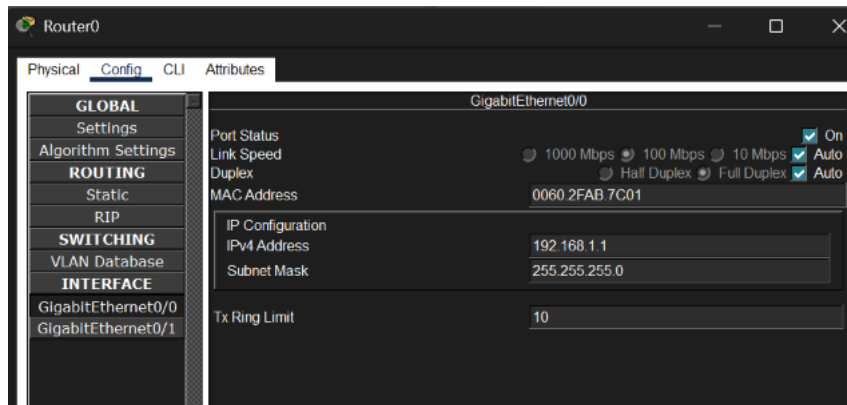
- 1) On the PC, go to the Desktop tab, open the Web Browser .
- 2) Enter the server's IP address in the URL bar to access the webpage hosted on the web server.

NETWORK TOPOLOGY DIAGRAM

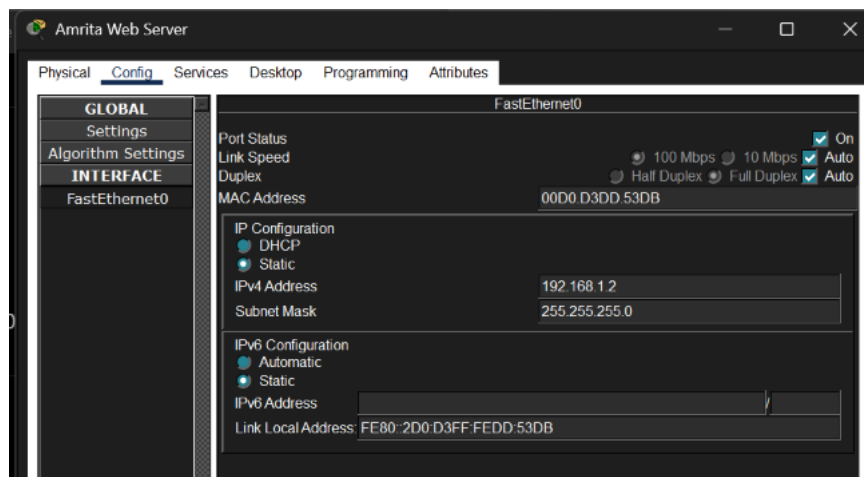


OUTPUT

1) Configure IP Address of Router:



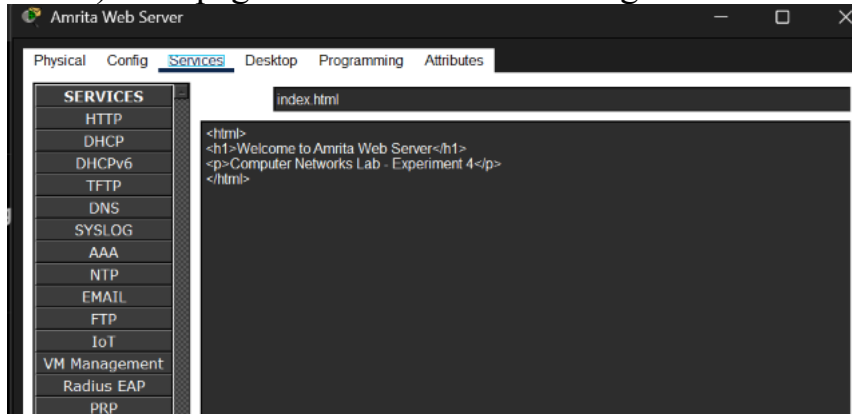
2) Configure IP address of Server:



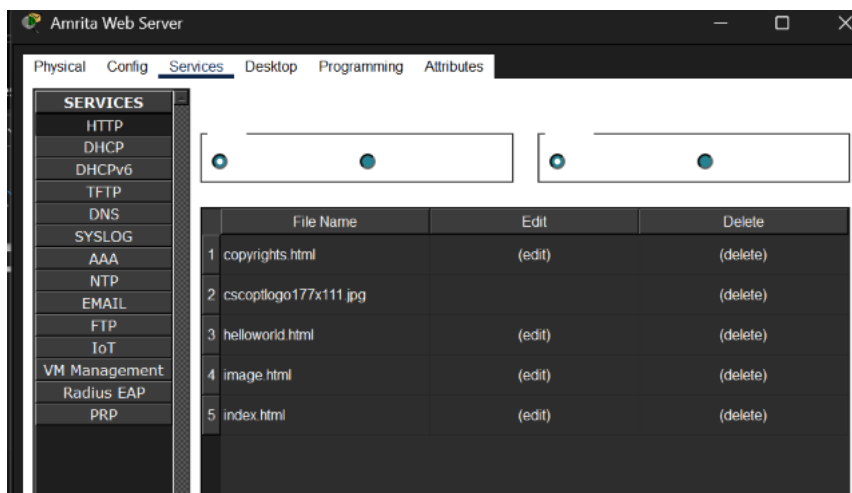
3) Configure the Router as DHCP Through Command Line Interface:

```
Router(config-if)#ip dhcp excluded-address 192.168.1.1 192.168.1.5
Router(config)#ip dhcp pool pool1
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#dns-server 192.168.1.10
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#exit
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

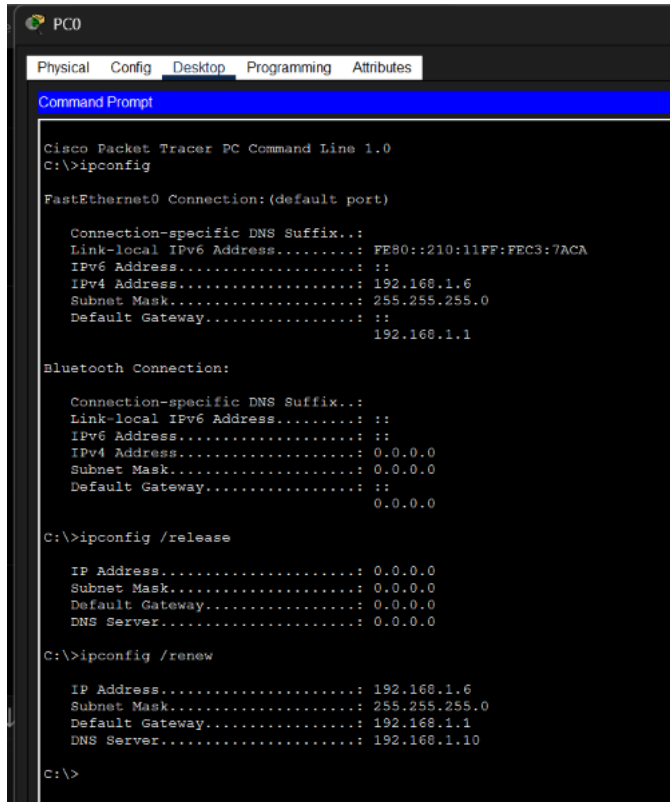
4) Webpage: Edit and save the changes made to the webpage



5) Web Server Configuration:



6) Verify the Configuration and obtain IP address on any PC:



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::210:11FF:FEC3:7ACA
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.6
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ipconfig /release

    IP Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0
    DNS Server . . . . .: 0.0.0.0

C:\>ipconfig /renew

    IP Address . . . . .: 192.168.1.6
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.1.1
    DNS Server . . . . .: 192.168.1.10

C:\>
```

7) Verify the web server access:



RESULT

We successfully configured a Cisco router to act as a DHCP server. The router dynamically assigned IP addresses to the connected devices within the defined IP range, and implemented a web server to access a webpage from connected devices within the network, ensuring efficient IP address management in the network.

1. First IP Configuration: obtain an IP address automatically via DHCP for Faculty PC.

The client PCs successfully obtained IP addresses automatically from the router configured as a DHCP server.

2. Release of IP address back to the DHCP pool using **ipconfig /release** and re-establish network connectivity to get a new IP address using **ipconfig /renew**:

The IP addresses were successfully released using `ipconfig /release` and renewed using `ipconfig /renew`.

3. Access of webpage from the connected devices in the network:

The webpage hosted on the server was successfully accessed from the client PCs through the web browser.

INFERENCE

Configuring a Cisco router as a DHCP server and implementing a web server within a network using Cisco Packet Tracer highlights essential aspects of dynamic IP address allocation and web service accessibility.

This setup involves multiple end devices, including five PCs and a server, connected through a switch to a 2911 router, demonstrating scalable network architecture. The router's DHCP configuration ensures that all connected devices receive IP addresses dynamically, streamlining network management and reducing manual configuration errors. By connecting the server to the switch and configuring it as a web server, the network allows all devices to access a webpage hosted on the server, showcasing integrated network services. This experiment underscores the importance of combining DHCP for efficient IP management with web server implementation for providing network services, illustrating the practical application of these concepts in managing and utilizing network resources effectively.

The router successfully functioned as a DHCP server by dynamically assigning IP addresses to all connected PCs, while the server provided

web services through HTTP. This experiment demonstrated the integration of DHCP and web server services in a local area network, enabling automatic IP allocation and successful webpage access.

Lab Scenario: Dynamic IP Allocation for a University Department Network

Scenario Title: *"Automating IP Address Assignment in the Computer Science Department"*

Background:

The **Computer Science Department** of a university has recently upgraded its lab infrastructure. There are **50 new client computers** that need to connect to the university network. The network administrator wants to avoid manually configuring IP addresses on each machine and instead use a **DHCP (Dynamic Host Configuration Protocol) server** to automatically assign IP addresses, subnet masks, default gateways, and DNS server details.

You have been assigned the task of **setting up and testing a DHCP server** in this environment.

Objective:

- Configure a DHCP server on a designated machine or router.
- Ensure that all client machines in the network receive dynamic IP configuration.
- Test and verify successful communication between clients and the network.

Lab Tasks:

Task 1: DHCP Server Configuration

1. Set up a DHCP server using either:
 - A dedicated DHCP service (e.g., on a Linux server), **OR**
 - A simulated router in **Cisco Packet Tracer**.
2. Configure the DHCP server with:

- **IP address range (scope):** 192.168.10.100 to 192.168.10.150
- **Subnet mask:** 255.255.255.0
- **Default gateway:** 192.168.10.1
- **DNS server:** 8.8.8.8

Q1: Provide a screenshot or console output showing the configured DHCP scope.

Task 2: Client Setup and Verification

1. Configure at least **three client PCs** to obtain IP addresses automatically (DHCP).

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.99
Router(config)#
Router(config)#ip dhcp pool CS_DEPT
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

The DHCP scope was successfully configured with the IP range **192.168.10.100–192.168.10.150**, subnet mask **255.255.255.0**, default gateway **192.168.10.1**, and DNS server **8.8.8.8**.

2. Check and record the IP addresses received by the clients.

Q2: What IP address, subnet mask, gateway, and DNS were assigned to each client?

Client	IP Address	Subnet Mask	Default Gateway	DNS
PC0	192.168.10.100	255.255.255.0	192.168.10.1	8.8.8.8
PC1	192.168.10.101	255.255.255.0	192.168.10.1	8.8.8.8
PC2	192.168.10.102	255.255.255.0	192.168.10.1	8.8.8.8

3. Test the connection between clients using ping.

Q3: Were the clients able to successfully communicate with each other and the gateway?

```
C:\>ping 192.168.10.101

Pinging 192.168.10.101 with 32 bytes of data:

Reply from 192.168.10.101: bytes=32 time<1ms TTL=128
Reply from 192.168.10.101: bytes=32 time<1ms TTL=128
Reply from 192.168.10.101: bytes=32 time<1ms TTL=128
Reply from 192.168.10.101: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.102

Pinging 192.168.10.102 with 32 bytes of data:

Reply from 192.168.10.102: bytes=32 time<1ms TTL=128
Reply from 192.168.10.102: bytes=32 time<1ms TTL=128
Reply from 192.168.10.102: bytes=32 time=11ms TTL=128
Reply from 192.168.10.102: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```

Yes. All the clients successfully communicated with one another and with the default gateway. The ping test was successful without packet loss

Task 3: Simulate DHCP Lease Expiry (Optional Extension)

1. Reduce the **lease time** to a short period (e.g., 1 minute).
2. Observe and document how and when the client requests a renewal.

Q4: What behavior did you observe when the lease expired? How did the client behave?

When the DHCP lease expired, the client automatically sent a renewal request to the DHCP server. After successful renewal, the client continued using the assigned IP address without interrupting network connectivity.

Task 4: Troubleshooting Scenario

A client PC is not receiving an IP address from the DHCP server. Diagnose and resolve the issue.

Q5: What were the possible causes, and how did you fix them?

Possible Causes:

- DHCP service disabled
- Wrong cable connection
- Router interface shutdown
- Incorrect DHCP pool configuration

- Client configured with Static IP
- Physical link down

How it was fixed:

The issue was resolved by enabling the router interface, verifying the DHCP pool configuration, ensuring the client was configured for DHCP, and checking that all network connections were active. After the corrections, the client successfully received an IP address from the DHCP server.